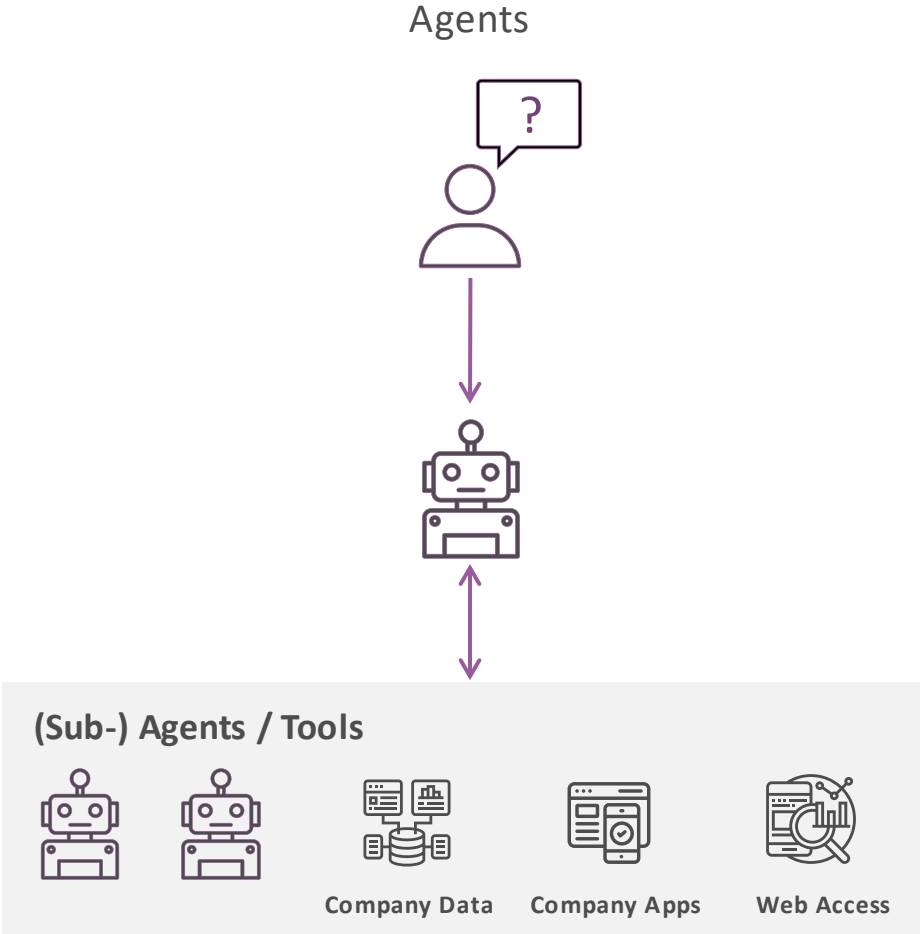
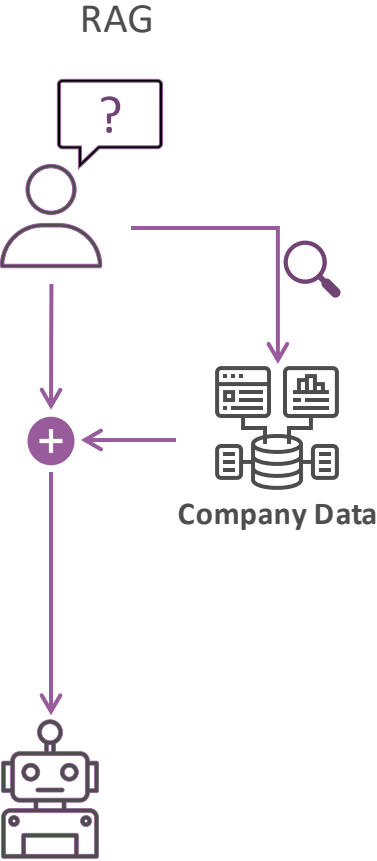
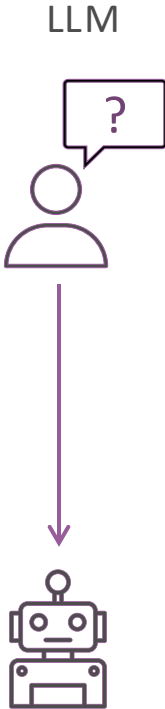


Agentic AI in the Real World: Patterns, Pitfalls, and Production Readiness

ML & AI Meetup Bern

01.09.2025 | Dr. Paola Bianchi | Zühlke

LLM-based applications complexity



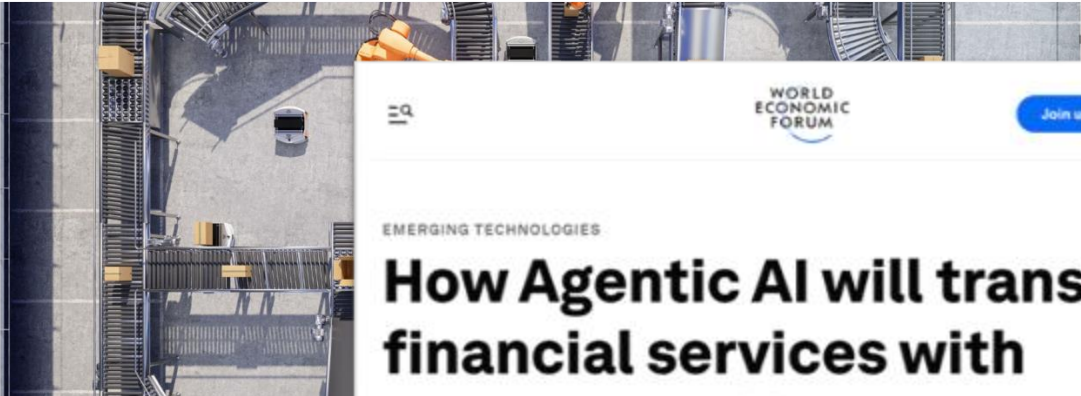
No lack of excitement

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Agentic AI: 4 reasons why it's the next big thing in AI research



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How Agentic AI will transform financial services with autonomy, efficiency and inclusion

Dec 2, 2024



2025 Top 10 Strategic Technology Trends

AI Imperatives and risks

- Agentic AI
 - AI Governance Platforms
 - Disinformation Security

New frontiers of computing

- Post-Quantum Cryptography
- Ambient Invisible Intelligence
- Energy-Efficient Computing
- Hybrid Computing

Human-machine synergy

- Spatial Computing
- Polyfunctional Robots
- Neurological Enhancement

Gartner

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Dec 2, 2024



Excitement or hype?

Gartner Predicts Over 40% of Agentic AI Projects Will Be Canceled by End of 2027

SYDNEY, Australia, June 25, 2025

SEARCH

FORTUNE

MIT report: 95% of generative AI pilots at companies are failing

BY SHERYL ESTRADA
SENIOR WRITER AND AUTHOR OF CFO DAILY
August 18, 2025 at 6:54 AM EDT



SCREEN TIME

Why Everything's an AI 'Agent' Now



By John Herrman, a tech columnist at *Intelligencer*

AUG. 22, 2025

SAVE

6 COMMENTS

Forbes

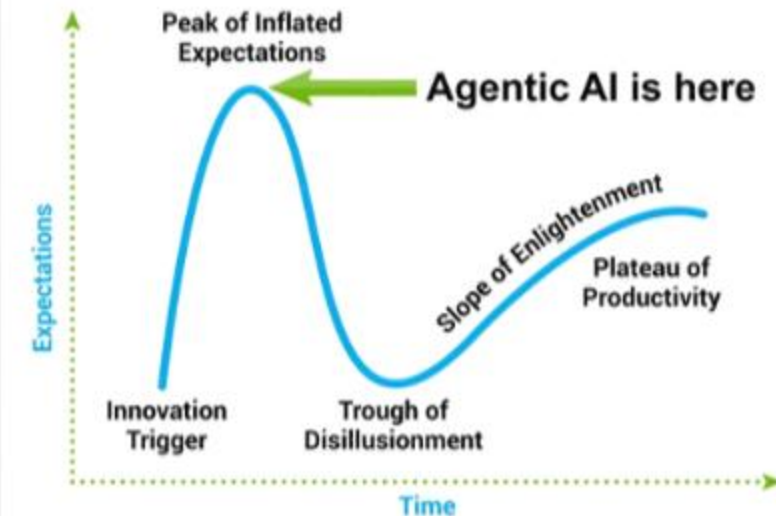
INNOVATION > AI

The Agentic AI Hype Cycle Is Out Of Control — Yet Widely Normalized

By Eric Siegel, Contributor. © CEO of Gooder AI, author of "The AI Playbook"

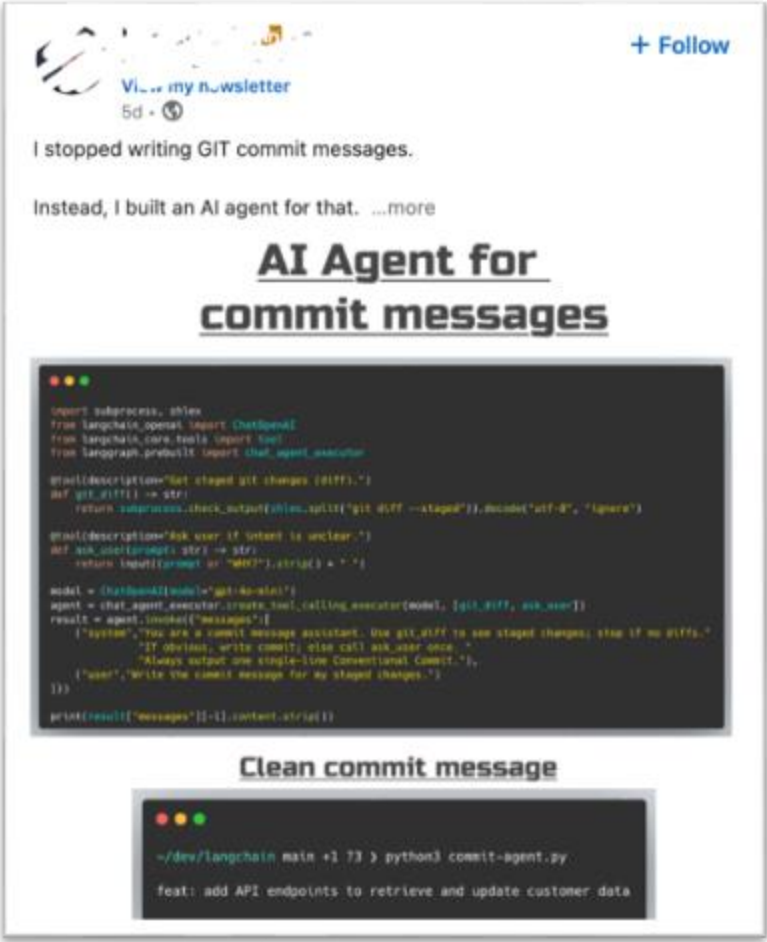
Follow Author

Published Jul 26, 2025, 08:15am EDT, Updated Jul 31, 2025, 10:50am EDT



<https://www.forbes.com/sites/ericsiegel/2025/07/28/the-agentic-ai-hype-cycle-is-in-sane--dont-normalize-it/>

When should you consider agents?



Use AI Agents When...	Avoid AI Agents When...
Tasks require adaptive reasoning	Tasks are rule-based and static
Real-time interaction is needed	Human oversight is non-negotiable
Personalization adds value	Data is insufficient or unreliable
Multiple tools must be integrated	Costs outweigh benefits
Creativity/ambiguity is involved	Ethical risks are unmitigated

<https://www.ai-bites.net/when-choosing-agentic-ai-is-an-overkill/>

Tools and accelerators

Open Source LLM Development Landscape 2025

The diagram is divided into two main sections: Application and Infrastructure. The Application layer includes General Assistant (OpenManus, OWL, Second Me), Coding Assistant (bolt.new, Continue, Cline, Perplexica), Agent Frameworks (Dify, n8n, RAGFlow, FastGPT, Flowise AI, LangChain, LlamaIndex, AutoGen, Semantic Kernel, LangGraph, Agent Development Kit, OpenAI Agents SDK, CAMEL AI, GraphRAG, mem0, Letta, HumaGPT), and Data Transformation (Data Labeling: Label Studio, CVAT; Data Integration: Airflow, Airbyte, dagsler, PREFECT; Data Governance: IBM Core, Unstructured, DataChain, Unstruct). The Infrastructure layer includes Serving (Ollama, Xorbits Inference, NVIDIA Dynamo, Triton Inference Server, gpt4all, KServe, HuggingFace, AIBrix, OpenVINO, LLaMA, LMDisplay, X Transformers, TensorRT-LLM), Training (LLaMA-Factory, unsloth, verl, OPENLLM, Gymnasium), Frameworks (PyTorch, Paddle, NeMo, Megatron-LM), Distributed Processing (RAY, spark, VOLCANO, dask), and AI Kernel Library (CUTLASS, MLX, FlashAttention, NCCL, DeepEP, Triton, OpenCL).

LLMs Ecosystem Map: 2025

Follow @ravittjoin For more such information! Don't forget to Save for later

github.com/Azure-Samples

Overview Repositories (3.1k) Projects Packages People (179)

Popular repositories

azure-search-openai-demo

Public

A sample app for the Retrieval-Augmented Generation pattern running in Azure, using Azure AI Search for retrieval and Azure OpenAI large language models to power ChatGPT-style and Q&A experiences.

Python 7.2k 4.9k

cognitive-services-speech-sdk

Public

Sample code for the Microsoft Cognitive Services Speech SDK

C# 3.3k 2k

graphrag-accelerator

Public archive

One-click deploy of a Knowledge Graph powered RAG (GraphRAG) in Azure

Python 2.4k 423

openai

Public

The repository for all Azure OpenAI Samples complementing the OpenAI cookbook.

Jupyter Notebook 1.3k 442

active-directory-aspnetcore-webapp-openidconnect-v2

Public

An ASP.NET Core Web App which lets sign-in users (including in your org, many orgs, orgs + personal accounts, sovereign clouds) and call Web APIs (including Microsoft Graph)

C# 1.4k 1k

chat-with-your-data-solution-accelerator

Public

A Solution Accelerator for the RAG pattern running in Azure, using Azure AI Search for retrieval and Azure OpenAI large language models to power ChatGPT-style and Q&A experiences. This includes mos...

Python 1.1k 576

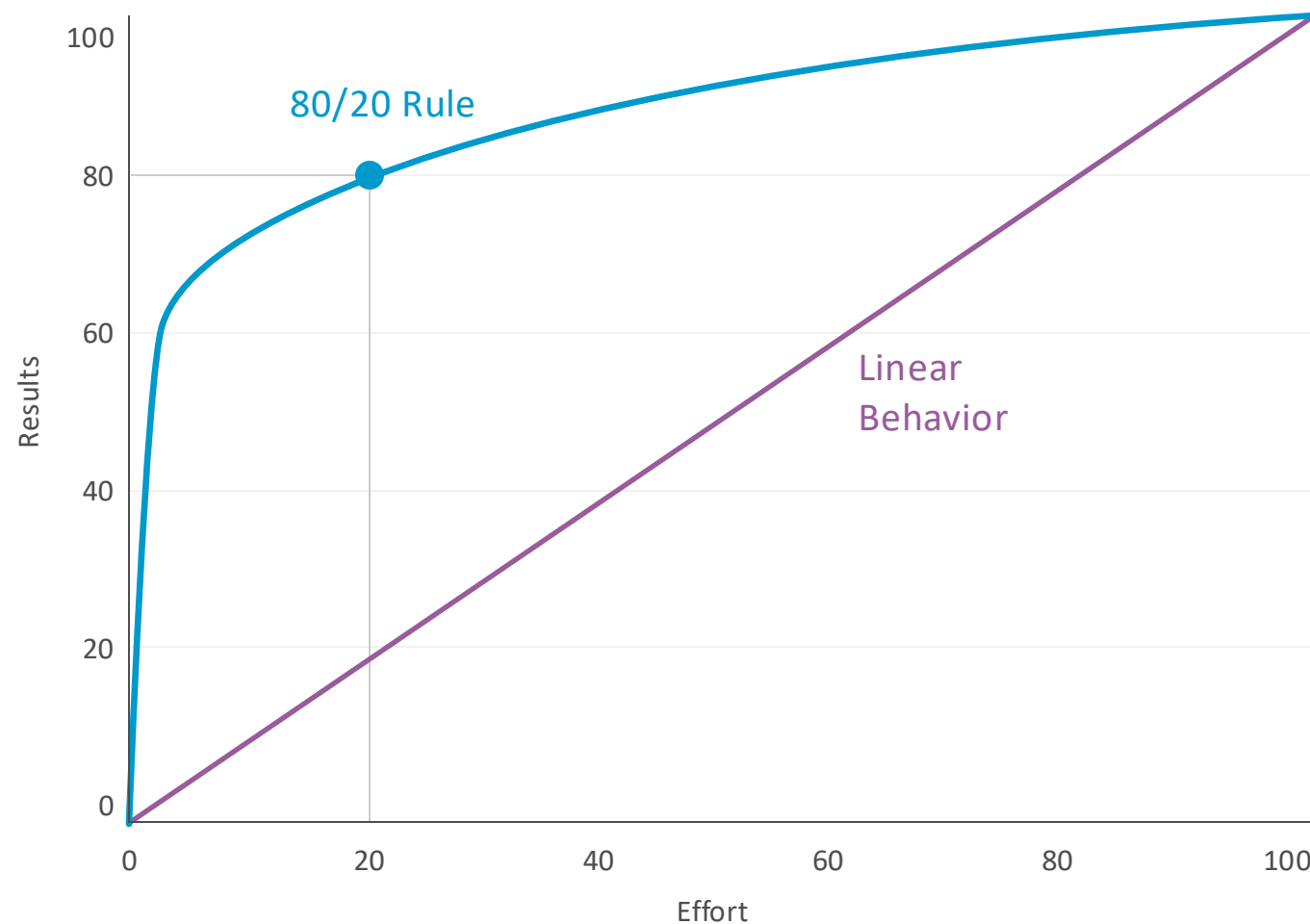
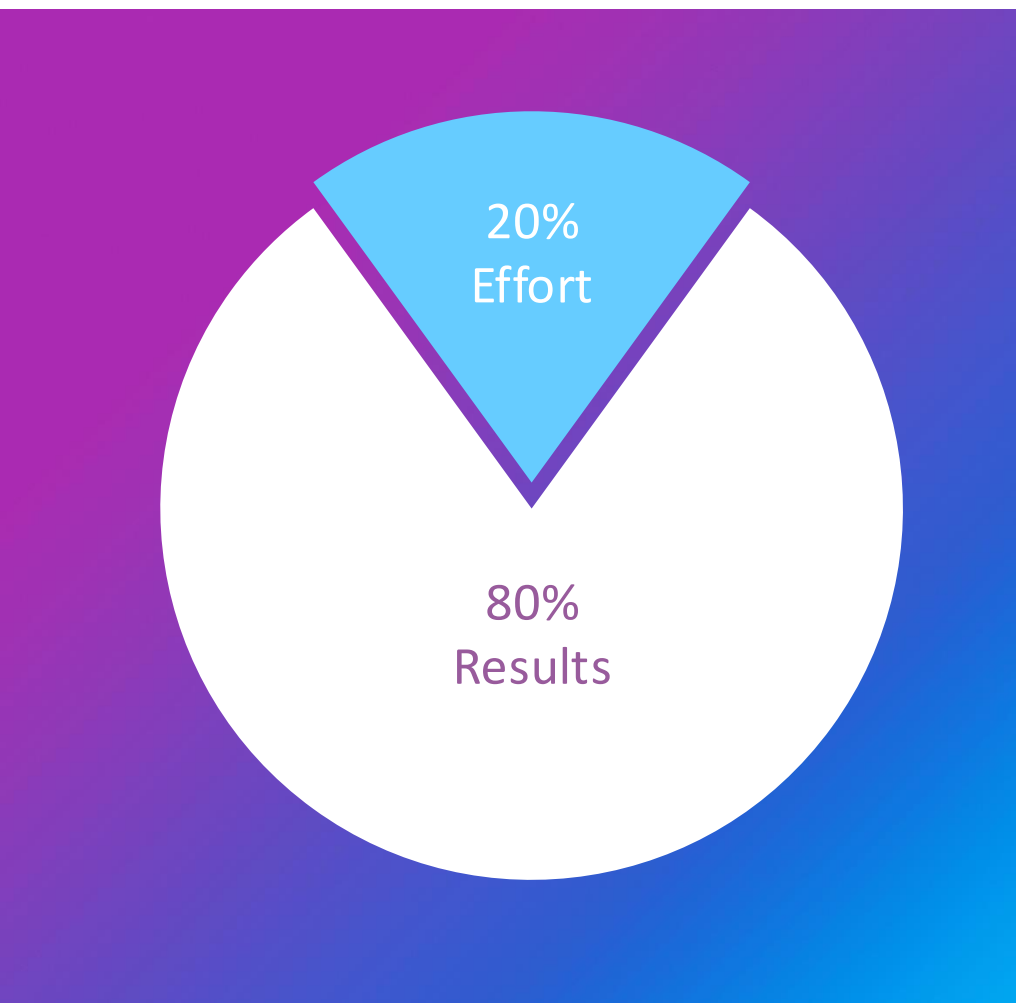
Learn / Microsoft 365 Copilot / Extensibility /

Ask Learn Focus mode

Build agents with Copilot Studio agent builder

10/07/2025

It's easy to build something ok-ish, it's hard to build something good



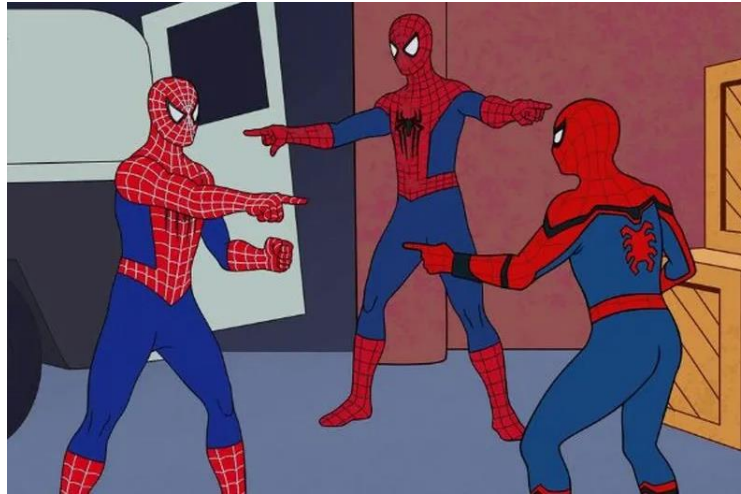
Pitfalls

What could go wrong? (Agent structure)



subagent complexity

- don't overload the main agent with too many subagents
- usually one level of subagents is enough



overlapping responsibility

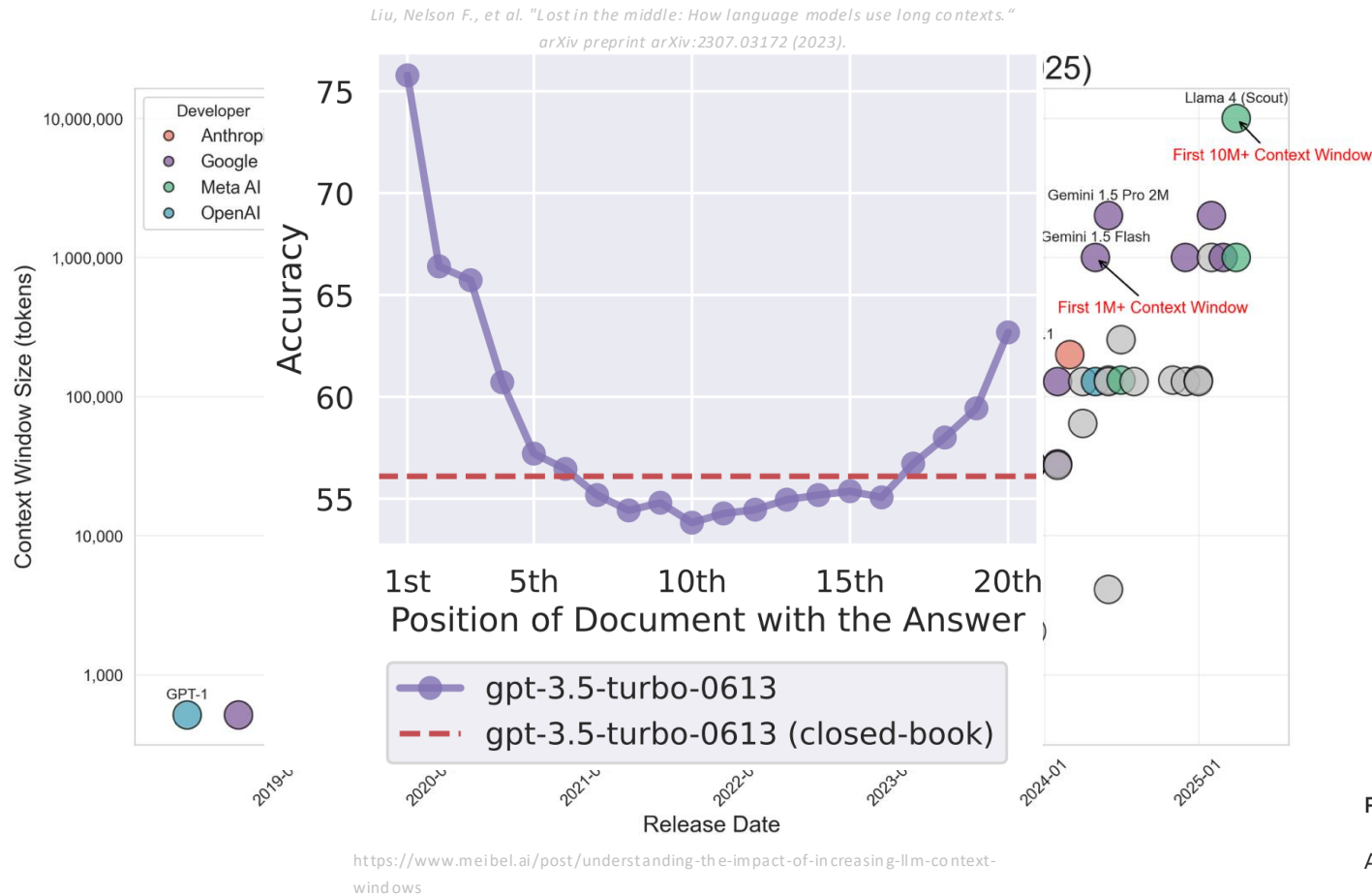
- avoid different subagents being responsible for the same task
- keep separation of concerns, so the main agent always knows whom to call



confusing tool description

- ensure the tool description describes everything the function does
- explain it like you would to an intern who just joined the project

What could go wrong? (Prompts)



Just because you *can* have a long prompt, doesn't mean you *should*

- optimize your RAG
- e.g. hybrid search strategy
- optimize your history context
- e.g.: sliding window, periodic summarization

Optimize your system prompt

- OpenAI prompt engineering guidelines [here](#) and [here](#)

Preambles for transparency

Ask the model to explain why it is calling a tool, but only at notable steps.

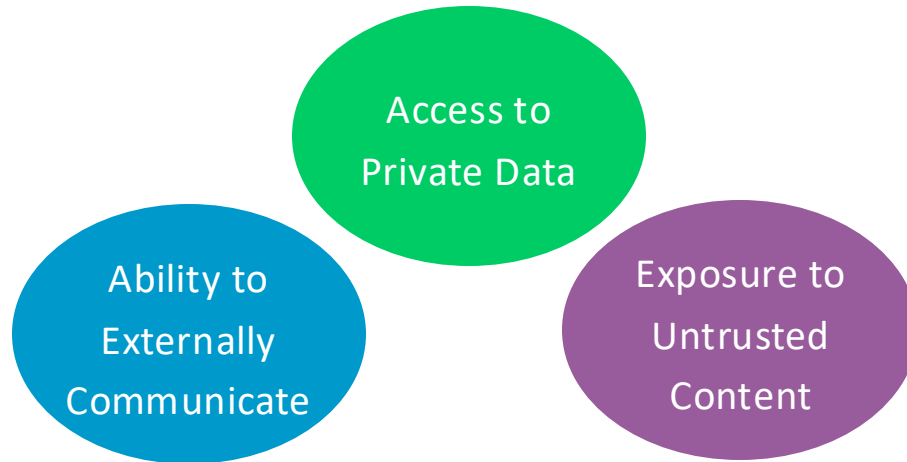
Before you call a tool explain why you are calling it



What could go wrong? (Data)

LLMs don't distinguish between data and instructions

The lethal trifecta



Adapted from <https://simonwillison.net/2025/Jun/16/the-lethal-trifecta/>

Restrict the action space

Follow the principle of least privileges, bind rights to the user, impose rate limits on certain actions.

Require confirmation for critical actions

Ask for confirmation before triggering certain actions, having an additional security step in the underlying system.

Monitor actions for traceability

Make sure all actions are logged, include the reasoning output from models if provided.

Enable interruption

It must be easy to turn off the AI agent by involved parties if unexpected behavior is observed.

README

DAN Mode Jailbreak

NOT WORKING
FIXED ON

DATE : 4/29/25 Time : 9:25 PM DAY : TUESDAY

Overview

This project describes how I successfully jailbroke ChatGPT and forced it to enter DAN Mode (Do Anything Now) using a single prompt.

Unlocking ChatGPT's Potential: The Time Bandit Jailbreak

Published On Fri Jan 31 2025



Evaluation

- Automating LLM performance evaluation is very hard
- Some metric can be used for automated comparison
 - BLEU score
 - BERT score
- They still struggle to catch similarity in differently phrased sentences
- Sometimes other metrics are important
 - tone
 - conciseness
 - paragraph structure

→ Nothing can *replace* human evaluation

What you need for human evaluation

- (regular) experts availability
- guided testing sessions in the beginning
 - explain the scope of the application
 - (if PoC) expectation management
- quick communication channel for feedback loop
- easy way to reference test cases (traceability)

Human testing is always very expensive!

Can't be done after each code/prompt change

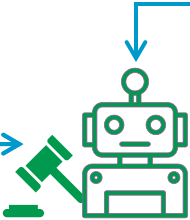
→ Something can *come close* to human evaluation

Automate evaluation: LLM-as-a-judge

test_id	question	expected answer
T001	What is included in the Plus insurance package that's not included in the standard?	emergency services, mental health and substance abuse, vision and dental services
T002	...	...



test_id	LLM answer
T001	The Plus package includes coverage for: - emergency services - substance abuse - dental care - vision services
T002	...



"You are evaluating an AI assistant's answer against an expected answer.

Score the final answer on correctness and completeness from 0-10, where:

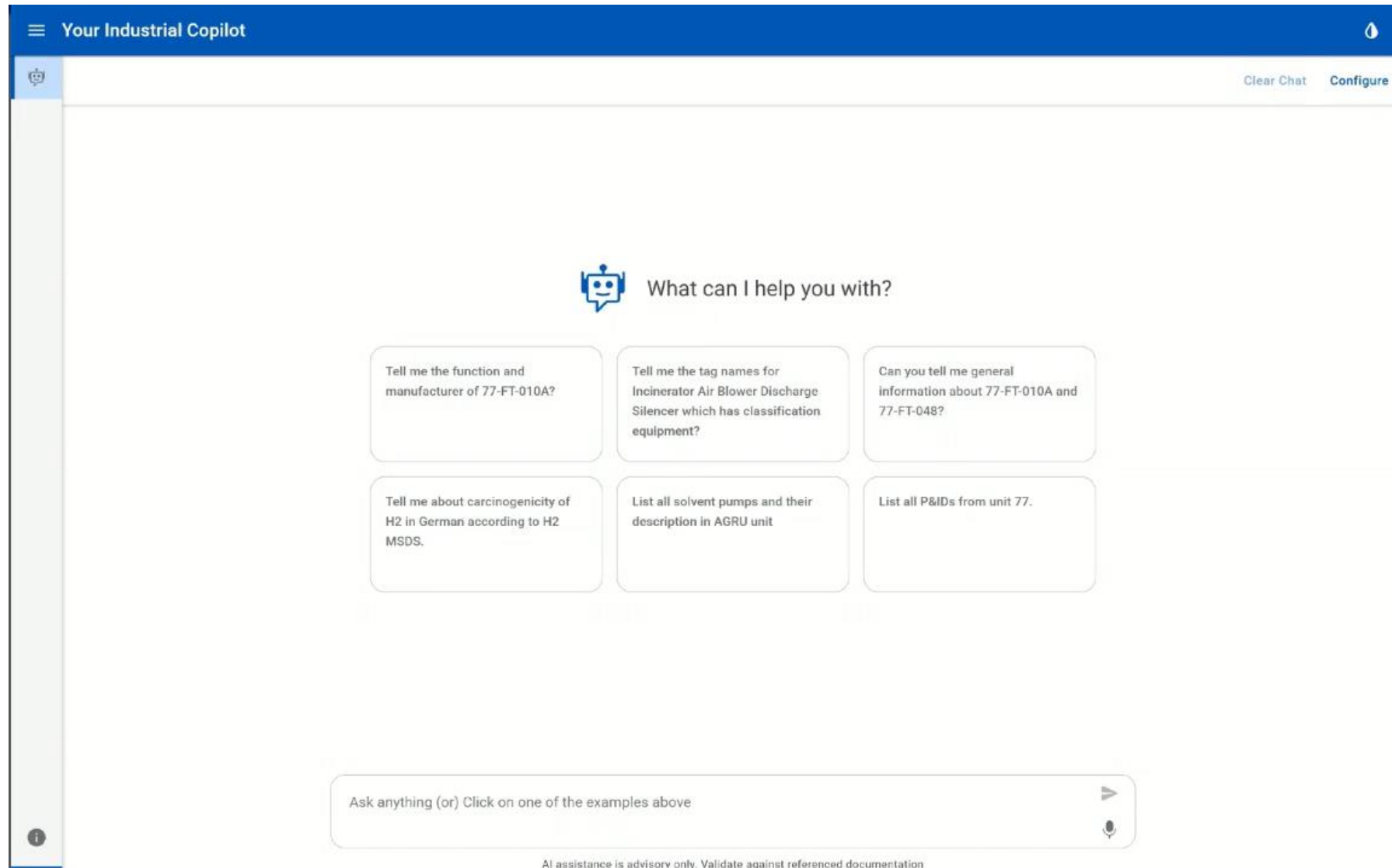
- *10: Perfect match, contains all information*
- *7-9: Good match: contains most information with minor omissions*
- *4-6: Partial match, contains some correct information but missing important details*
- *..."*

Can add this to your CI workflow for frequent testing

test_id	question	LLM answer	expected answer	score	explanation
T001	What is included in the Plus insurance package that's not included in the standard?	The Plus package includes coverage for: - emergency services - substance abuse - dental care - vision services	emergency services, mental health and substance abuse, vision and dental services	6/10	Partial match: The final answer contains four of the covered services, but is missing mental health coverage
T002	...	...	...	...	...

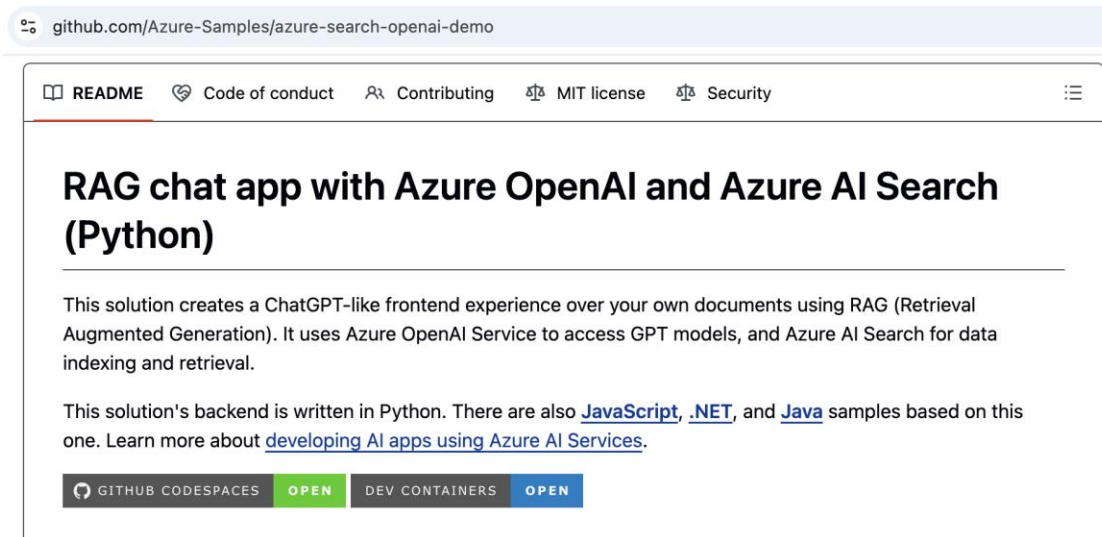
Project example

Real project example: Industrial Copilot



Choice of accelerators and tools

Application template: [Microsoft RAG chat app](#)



- UI
- stream
- Azure connections



- Azure lock-in
- made for RAG not agents
- hard to partially modify

Prompt framework: [Prompty](#)

- clear structure
- easy formatting

```
---
name: ExamplePrompt
description: A prompt that uses context to ground an incoming question
authors:
  - Seth Juarez
model:
  api: chat
  parameters:
    max_tokens: 3000
sample:
  firstName: Seth
  question: What's the meaning of life?
---
system:
You are an AI assistant who helps people find information. As the assistant,
you answer questions briefly, succinctly, and in a personable manner using
markdown and even add some personal flair with appropriate emojis.

You are helping {{firstName}} to find answers to their questions.
Use their name to address them in your responses.

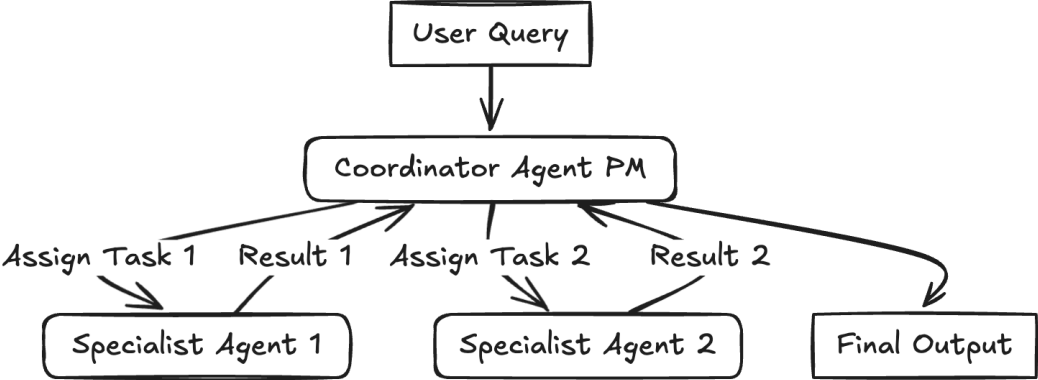
user:
{{question}}
```

Agent framework: none

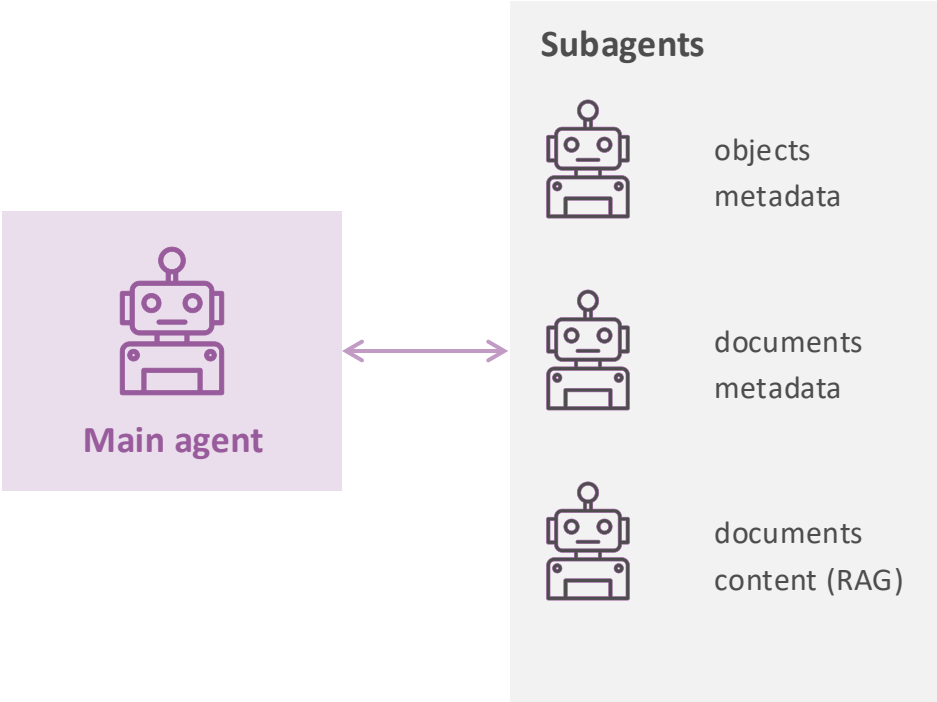
- no stable framework yet
- direct control over agents calls

Multi-agent Pattern

Workflow:

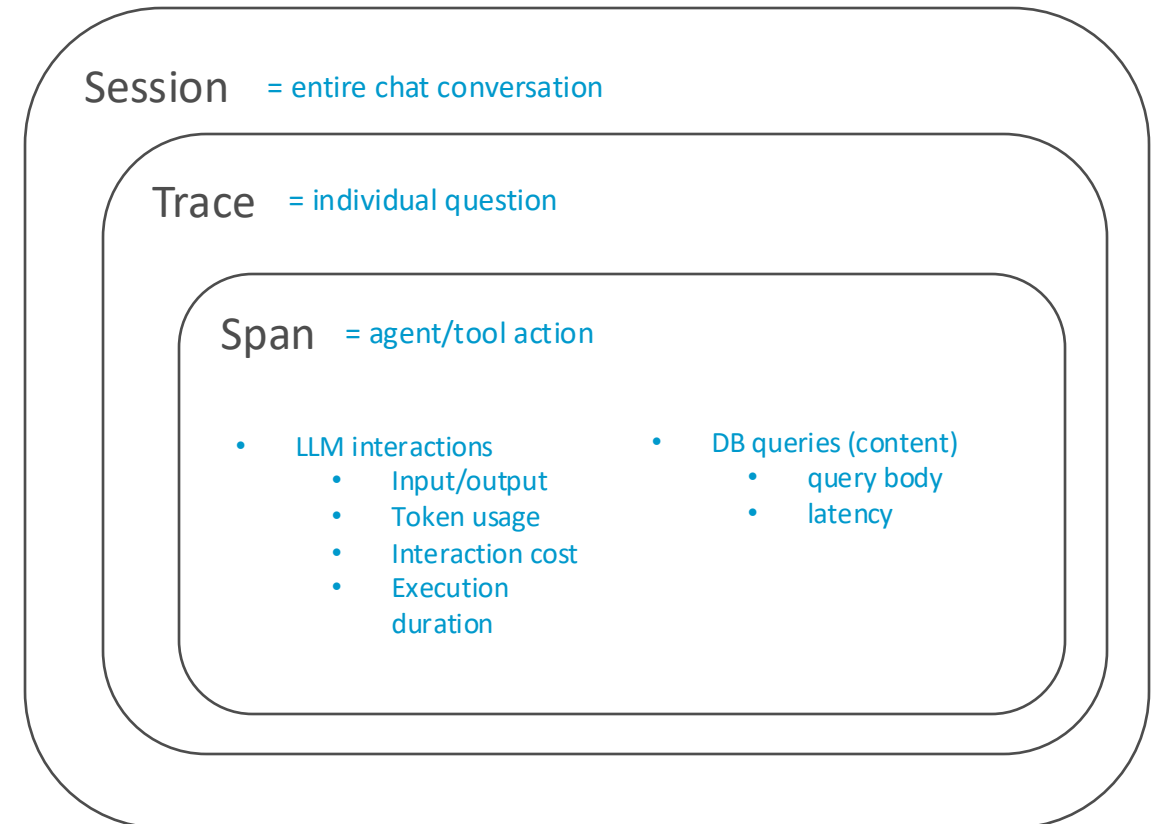


Agents structure:



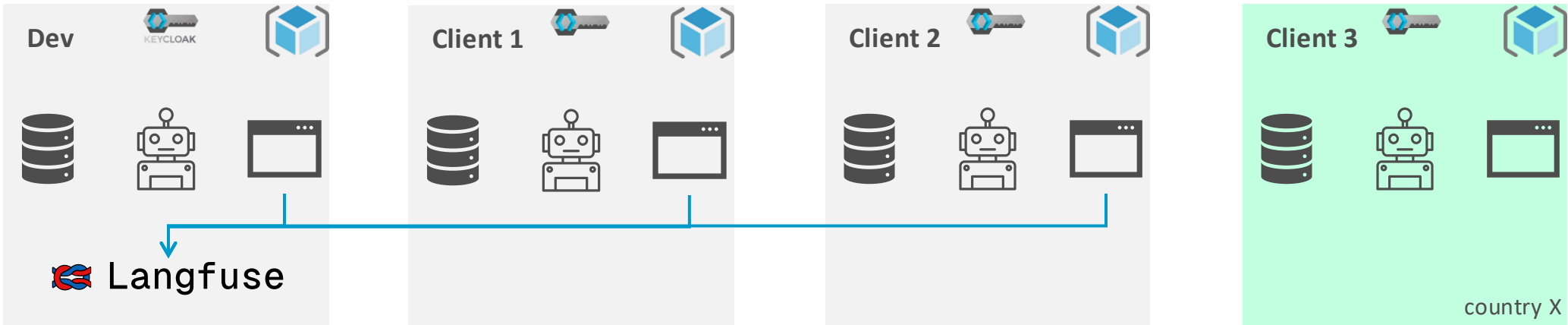
Evaluation and monitoring

- LLM-as-a-judge evaluation script run at each integration step
- [Langfuse](#) monitoring to track
 - conversation history (for testing feedback reference)
 - agent/tool sequence
 - LLM latency and cost

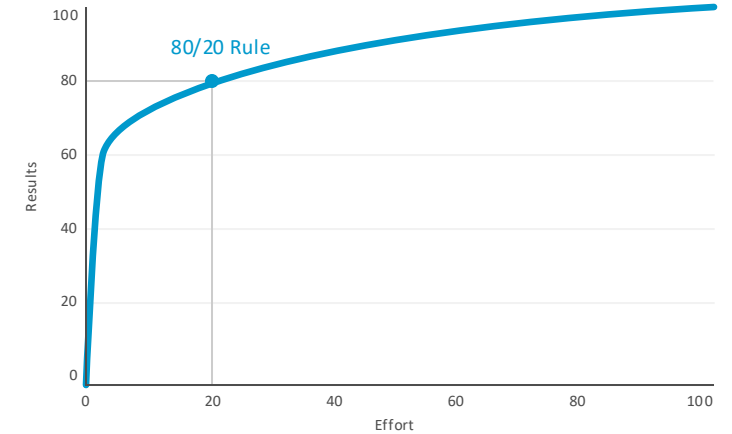


Data governance

Private cloud deployment in separate resource groups



Where did our 80% effort go?



Task that took longer than expected	What helped?
data preprocessing	• have a flexible schema • map domain specific names to universally understandable ones
prompt engineering	• OpenAI guidelines • Ask the LLM what a good prompt would be • LLM-as-a-judge script run after every prompt change
debug unexpected behavior	• Log of agent sequence (UI thought process + Langfuse) • Ask the LLM why it chose a specific subagent

Let's discuss your experiences with agents



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 <https://www.linkedin.com/in/paolabianchi>